

- 1(a) Zac is planning to write a report on the music preferences of the students at his college. There is a large number of students at the college.

State one reason why Zac might wish to obtain information from a sample of students, rather than from all the students. [1]

- (b) Amaya suggests that Zac should use a sample that is stratified by school year.

Give one advantage of this method as compared with random sampling, in this context. [1]

- (c) Zac decides to take a random sample of 60 students from his college. He asks each student how many hours per week, on average, they spend listening to music during term. From his results he calculates the following statistics.

Mean	Standard deviation	Median	Lower quartile	Upper quartile
21.0	4.20	20.5	18.0	22.9

Sundip tells Zac that, during term, she spends on average 30 hours per week listening to music.

Discuss briefly whether this value should be considered an outlier. [3]

- (d) Layla claims that, during term, each student spends on average 20 hours per week listening to music. Zac believes that the true figure is higher than 20 hours. He uses his results to carry out a hypothesis test at the 5% significance level.

Assume that the time spent listening to music is normally distributed with standard deviation 4.20 hours.

Carry out the test.

[7]

2(a) In the past, the time for Jeff's journey to work had mean 45.7 minutes and standard deviation 5.6 minutes. This year he is trying a new route. In order to test whether the new route has reduced his journey time, Jeff finds the mean time for a random sample of 30 journeys using the new route. He carries out a hypothesis test at the 2.5% significance level.

Jeff assumes that, for the new route, the journey time has a normal distribution with standard deviation 5.6 minutes.

State appropriate null and alternative hypotheses for the test.

[2]

(b) Determine the rejection region for the test.

[4]

- 3 The level, in grams per millilitre, of a pollutant at different locations in a certain river is denoted by the random variable X , where X has the distribution $N(\mu, 0.000\ 040\ 9)$.

In the past the value of μ has been 0.0340.

This year the mean level of the pollutant at 50 randomly chosen locations was found to be 0.0325 grams per millilitre.

Test, at the 5% significance level, whether the mean level of pollutant has changed.

[7]

4 Research has shown that drug A is effective in 32% of patients with a certain disease.

In a trial, drug B is given to a random sample of 1000 patients with the disease, and it is found that the drug is effective in 290 of these patients.

Test at the 2.5% significance level whether there is evidence that drug B is effective in a lower proportion of patients than drug A.

[7]

- 5 In the past, the time spent by customers in a certain shop had mean 10.5 minutes and standard deviation 4.2 minutes. Following a change of layout in the shop, the mean time spent in the shop by a random sample of 50 customers is found to be 12.0 minutes.
- (a) Assuming that the standard deviation is unchanged, test at the 1% significance level whether the mean time spent by customers in the shop has changed. [7]

- (b) Another random sample of 50 customers is chosen and a similar test at the 1% significance level is carried out. Given that the population mean time has not changed, state the probability that the conclusion of the test will be that the population mean time **has** changed. [1]

- 6 A researcher is investigating the actual lengths of time that patients spend at their appointments with the doctors at a certain clinic. There are 12 doctors at the clinic, and each doctor has 24 appointments per day. The researcher plans to choose a sample of 24 appointments on a particular day.

(a) The researcher considers the following two methods for choosing the sample.

Method A: Choose a random sample of 24 appointments from the 288 on that day.

Method B: Choose one doctor's 1st and 2nd appointments. Choose another doctor's 3rd and 4th appointments and so on until the last doctor's 23rd and 24th appointments.

For each of A and B state a disadvantage of using this method.

[2]

Appointments are scheduled to last 10 minutes. The researcher suspects that the actual times that patients spend are more than 10 minutes on average. To test this suspicion, he uses method A, and takes a random sample of 24 appointments. He notes the actual time spent for each appointment and carries out a hypothesis test at the 1% significance level.

(b) Explain why a 1-tail test is appropriate.

[1]

The population mean of the actual times that patients spend at their appointments is denoted by μ minutes.

(c) Assuming that $\mu = 10$, state the probability that the conclusion of the test will be that μ is not greater than 10.

[1]

The actual lengths of time, in minutes, that patients spend for their appointments may be assumed to have a normal distribution with standard deviation 3.4.

(d) Given that the total length of time spent for the 24 appointments is 285 minutes, carry out the test. [7]

(e) In part (d) it was necessary to use the fact that the sample mean is normally distributed. Give a reason why you know that this is true in this case. [1]

- 7 The times taken by employees to complete a task are normally distributed with standard deviation 2.6 minutes. A manager claims that the mean time is 15.5 minutes but an employee suspects that the mean time is greater than this. He intends to carry out a hypothesis test at the 5% significance level to test this claim. He records the times taken by a random sample of 12 employees.

(a) Find the critical region for the test.

[3]

(b) The total of the times taken by the 12 employees was 202.1 minutes. Carry out the test.

[5]

- 8 In the past, the time spent in minutes, by customers in a certain library had mean 32.5 and standard deviation 8.2.
- Following a change of layout in the library, the mean time spent in the library by a random sample of 50 customers is found to be 34.5 minutes.
- Assuming that the standard deviation remains at 8.2, test at the 5% significance level whether the mean time spent by customers in the library has changed. [7]

- 9 It is known that under the standard treatment for a certain disease, 9.7% of patients with the disease experience side effects within one year. In a trial of a new treatment, 450 patients with this disease were selected and the number, X , that experienced side effects within one year was noted.

It was found that 51 of the 450 patients experienced side effects within one year.

- (a) Test, at the 10% significance level, whether the proportion of patients experiencing side effects within one year is greater under the new treatment than under the standard treatment.

[7]

- (b) It was later discovered that all 450 patients selected for the trial were treated in the same hospital. Comment on the validity of the model used in part (a).

[1]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1	a		Population large oe	B1 [1]	or, eg Would take too long to contact all students. NOT "Easier"
	b		eg: Includes students from all years (or ages) Numbers in years in correct proportions Different years might like different music	B1 [1]	or: Different years may have different numbers of students NOT "It's more representative" or "Takes all students into account" "You get a range of people""It avoids bias"
	c		$21 + 2 \times 4.2 = 29.4$ $22.9 + 1.5(22.9 - 18.0) = 30.25$ Unclear whether 30 is an outlier	B1 B1 B1 [3]	Allow "30 is more than 2 sds away from the mean" Allow "30 is less than $1.5 \times \text{IQR}$ from UQ" or eg "It depends which definition you use." Any comment implying uncertainty Ignore comments about mean ± 3 sds or mean ± 1 sd <u>Examiner's Comments</u> The specification (paragraph 2.02j) gives two definitions for an outlier. Most candidates used only one of these definitions, and so did not discover that they give different results. What was required for full marks was a recognition of the uncertainty about whether or not 30 should be considered an outlier. Candidates who used other definitions (such as $x > \mu + 3s$ or $x > \mu + s$) gained no credit. Some candidates did a calculation that mixed the median with the standard deviation or the mean with the IQR.

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
	d	H0: $\mu = 20$ H1: $\mu > 20$ where μ = pop mean time spent $\bar{X} \sim N(20, \frac{4.2^2}{60})$ and $\bar{X} = 21$ $P(\bar{X} > 21) = 0.0326$ Compare 0.05 Alternative methods for M1A1A1 or $\frac{a-20}{4.2/\sqrt{60}} = 1.645$ ($a = 20.9$) CV = 20.9 21 > 20.9 or 21 not in acceptance region or $\frac{21-20}{4.2/\sqrt{60}} (= 1.84)$ zcalc = 1.84 Compare 1.645 $\bar{X} \sim N(20, \frac{4.2^2}{60})$ and $\bar{X} = \frac{1260}{60}$ or 21 $P(\bar{X} < 21) \leq 0.9674$ Compare 0.95 Reject H_0 Condon Accept H_1 There is evidence that (mean) time spent is > 20 hours or eg there is evidence to support Zac's belief NB Use of $4.2^2/60$ as sd gives $p = 0.000335$	B1 B1 M1 A1 A1 Alternative methods for M1A1A1 M1 A1 A1 M1 A1 A1 M1 A1f [7]	NB Allow 2 sf throughout Allow other letters, not X unless defined. Not $\bar{X}$ B1B0 for 1 error eg 2-tail or: 2-tail B1B0 $\mu =$ B1B0 sample mean implied undefined B1B0 Not B1B0 μ include value 20 eg $H_0 = 20$ B1B0 etc: not in B1B0 terms of parameter Correct distribution and value of $\bar{X}$ stated or implied eg by 0.0326 or 0.967 or 20.9 or 1.84 or 0.000335 even if within incorrect statement eg $P(X = 21) = 0.0326$ Condone $\frac{4.2^2}{\sqrt{60}}$ or $\frac{4.2^2}{60^2}$ or $\frac{4.2}{60}$ BC Allow 2 sf, ie 0.033 Dep 0.0326 or 1.84 or 0.9674 or $P(X > 21)$ or $P(\geq 21)$ soi Must compare like with like, eg NOT prob of z-value or large prob of small prob or CV of wrong end of acceptance region Alternative methods for M1A1A1 Condone $\frac{4.2^2}{\sqrt{60}}$ or $\frac{4.2^2}{60^2}$ or $\frac{4.2}{60}$ Condone $\frac{4.2^2}{\sqrt{60}}$ or $\frac{4.2^2}{60^2}$ or $\frac{4.2}{60}$ Condone $\frac{4.2^2}{\sqrt{60}}$ or $\frac{4.2^2}{\sqrt{60}}$ or $\frac{4.2}{60}$ BC Dependent on clearly valid comparison of like with like. Dep 0.0485 or 0.951 or 20.9 or 1.84 or $P(X > 21)$ or $P(X \geq 21)$ soi May be implied by conclusion, eg "There is evidence that mean time is > 20 hours" M1A1

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
					In context, not definite; eg "Mean time is > 20 hours": A0 But "There is evidence to reject H0 and that mean time is > 20 h" Allow opposite conclusion, ft their values, if above conditions met and loses 2nd A1. But potentially can score all the other 6 marks Examiner's Comments  Misconception One difficult feature of hypothesis testing is that in some cases the calculated value of the statistic is significant when it is greater than the critical value, whereas in other cases the calculated value is significant when it is less than the critical value. In both these questions there was confusion about this. Many candidates used the sample mean, 21, as the mean of the distribution, rather than the suggested population mean of 20.
			Total	12	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
2	a		$H_0: \mu = 45.7,$ where μ = mean of all new journey times $H_1: x < 45.7$	B1 (AO1.1) B1 (AO2.5) [2]	Allow "where μ = mean journey time" Allow different letters Subtract use of B0B0 B1 for "p" unless each error defined eg: 2-tail B1B0 μ = sample mean implied B1B0 undefined B1B0 Not include value 45.7 B0B0 μ H0 = 45.7 B0B0 not in terms of parameter etc: <u>Examiner's Comments</u> As in Question 10, many candidates omitted to define μ
	b		$N(45.7, \frac{5.6^2}{30})$ and probability = 0.025 soi $P(\bar{X} < a) = 0.025$ $a = 43.7$ (3 sf) (43.696....) Rejection region is $(\bar{X}) < 43.7$ (3 sf) or $(\bar{X}) < 43.6$ with explanation 3.7 (3 sf)	M1* (AO3.3) M1 (AO1.1) A1f (AO1.1) A1 (AO1.1) [4]	or $N(45.7, \frac{392}{375})$ or $N(45.7, 1.045)$ and probability = 0.025 soi soi Dep M1 Allow $\leq$. Answer $(\bar{X}) < 34.7$ SC B1 (from not $\div$ by $\sqrt{30}$) Correct answer with inadequate or no working: SC B2 If (a) $\mu > 45.7$, allow $(\bar{X}) > 47.7$, M1M1A1A0
			Total	6	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
3			See the exemplars at the end of the MS Allow 2 sf throughout $H_0: \mu = 0.034$	B1(AO1.1) B1(AO2.5)	Subtract B1 for each error eg: 1-tail B1B0 undefined μ B1B0 not in terms of parameter B1B0 μ = sample mean implied B1B0 Not include value 0.034 B0B0 eg $H_0 = 0.034$ etc:	If 1-tail test: $H_0: \mu = 0.034$, defined μ , B1 $H_1: \mu < 0.034$ B0
			$H_1: \mu \neq 0.034$ where μ = (pop) mean pollutant level			
			Allow any letter (except X and $\bar{X}$: B0B0)			
			$N(0.034, \frac{0.0000409}{50})$ & $\bar{X} < 0.0325$ (condone $>, =$)	M1(AO3.3)		
			or $\frac{\bar{x} - 0.034}{0.00640 \div \sqrt{50}} = -1.96$			
			$P(\bar{X} < 0.0325) = 0.0486$ or CV is 0.0322 or acceptance region is 0.0322 to 0.0358 or $P(\bar{X} > 0.0325) = 0.951$	A1*(AO3.4)		
			0.0486 > 0.025 or 0.0325 > 0.0322 or 0.0325 is in AR or 0.951 < 0.975 A1A1	A1(AO1.1)	or 1.66 < 1.96 or -1.66 > -1.96 dep A1* Must be seen, allow on diag	$P(\bar{X} < 0.0325) = 0.0486$ or CV is 0.0325(1) A1
			Do not reject H_0 Or Insufficient (or No) evidence to reject H_0 Allow Accept H_0	M1(AO1.1)	Dep 0.0486 or 0.0322 or -1.66 or 0.951 < 0.975 seen, or $P(\bar{X} < 0.0325)$ stated or implied (possibly with wrong prob leading to opposite conclusion**) but 0 if 0.951 > 0.025	Comp 0.05 or ± 1.645 A1 or 0.0325 < 0.0325(1) or 0.0325 is within CR No more marks
						May be implied by conclusion
						Condone Reject H_1

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
			Insufficient (or No) evidence that (mean) pollutant level has changed oe or eg "It is unlikely that level has changed" "We can assume that level hasn't changed"	A1f(AO2. 2b) [7]	M0 In context. Context may be implied by eg "level" or "pollutant" ft only ** above Not "There is evidence that mean level has not changed" A0 Not definite, eg "Mean level has not changed " A0 <u>Examiner's Comments</u> Many candidates appeared to have been well-drilled in hypothesis test questions and gave very good answers. Some of these candidates lost one mark because they did not define the "μ" in their hypotheses. Others made arithmetical errors in their calculation of a probability or a critical value. Some other common errors were as follows. Misunderstanding the N(..., ...) notation, e.g. using a variance of 0.000 040 9². Failure to divide the variance by 50. H₁: $\mu = 0.0325$ or H₁: $\mu < 0.0340$. Use of 1.645 instead of 1.96 in finding the critical value or critical region, despite a correct H₁. Claiming that because 0.0486 > 0.025, H₀ should be rejected. Giving the final conclusion as a definite statement, such as "The mean pollutant level has not changed". Centres are referred to the exemplars at the end of the published mark scheme for further guidance.
			Total	7	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
4			$H_0: p = 0.32$ where $p = P(\text{drug B is effective})$ $H_1: p < 0.32$ $X \sim B(1000, 0.32)$ and $X = 290$ $P(X \leq 290) = 0.0221$ Comp 0.025 Reject H_0 There is evidence that drug B is effective in a lower prop of patients than drug A	B1 (AO 1.1) B1 (AO 2.5) M1 (AO 3.3) A1 (AO 3.4) A1 (AO 1.1) M1 (AO 1.1) A1FT (AO 2.2b) [7]	B1 for one error, eg undefined p , or $\neq$ soi BC In context. Not definite. Full statement	
			Total	7		
5		a	$H_0: \mu = 10.5$ where μ is pop mean time in shop $H_1: \mu \neq 10.5$ $\bar{X} \sim N(10.5, \frac{4.2}{\sqrt{50}})$ and $X = 12$ $P(\bar{X} > 12) = 0.00578$ or better Compare with 0.005 Do not reject H_0 Insufficient evidence that mean time has changed	B1 (AO1.1) B1 (AO2.5) M1 (AO3.3) A1 (AO3.4) M1 (AO1.1) M1 (AO1.1) A1 (AO2.2b) [7]	One error, eg undefined μ or 1-tail: B0B1 May be implied or 0.006 or 0.0058 BC In context. Not definite, eg "Mean time has not changed" A0	
		b	0.01	B1 (AO1.2) [1]		
			Total	8		

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
6		a	Method A: eg May not be representative, e.g. may have many (or few) appointments from one Dr Or may have many a.m. appts & few p.m. Method B: eg Time of 1st Dr's 1st & 2nd appointments may not be typical of his later ones. Similar for other Drs	B1(AO 3.5b) B1(AO 3.5b) [2]	Or may not have a good spread of appts Or other sensible Or other sensible	
		b	Researcher suspects "more than" 10 mins	B1(AO 2.4) [1]	oe	
		c	0.99	B1(AO 1.2) [1]		
		d	$H_0: \mu = 10$ where μ is pop mean appointment time $H_1: \mu > 10$ $\bar{X} \sim N(10, \frac{3.4}{\sqrt{24}})$ and $X = \frac{285}{24}$ (= 11.875) $P(\bar{X} > 11.875) = 0.00345$ or better Compare with 0.01 Reject H_0 Sufficient evidence that mean time is > 10 mins	B1(AO 1.1) B1(AO 2.5) M1(AO 3.3) A1(AO 3.4) M1(AO 1.1) M1(AO 2.2b) A1(AO 3.5a) [7]	One error, eg undefined μ or 2-tail: B0B1 May be implied or 0.003 BC In context. Not definite, eg "Mean time is > 10 mins": A0	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		e	Times spent assumed normally distributed, hence sample mean also normally distributed	B1(AO 1.2) [1]	oe
			Total	12	
7		a	$X \sim N(15.5, \frac{2.6^2}{12})$ $15.5 + 1.645 \frac{2.6}{\sqrt{12}}$ Critical region is $\bar{x} > 16.7$ (3 sf)	M1(AO 3.3) M1(AO 3.4) A1(AO 1.1) [3]	stated or implied
		b	$H_0: \mu = 15.5$ $H_1: \mu > 15.5$ where μ is mean time by all employees $\bar{x} = 16.8$ 16.8 is within CR ft their $\bar{x}$ & CR Reject H_0 . There is evidence that mean time for task is greater than 15.5 (minutes)	B1(AO1. 1) B1(AO2. 5) A1ft(AO3 .3) M1(AO1. 1) A1ft(AO2 .2b) [5]	In terms of parameter values B1B0 one error eg undefined μ or two-tail Use of 17.5 B0B0 OR $P(X > 16.8) = 0.0416$ (3 sf) Comp 0.05 Allow 0.25 if $H_1: \mu \neq 15.5$ In context, not definite. ft their 0.0416 but not comp with 0.25
			Total	8	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
8			$H_0 : \mu = 32.5$ $H_1 : \mu \neq 32.5$ where μ is mean time spent by all customers $X \sim N\left(32.5, \frac{8.2^2}{50}\right)$ and $X > 34.5$ $P(X > 34.5) = 0.0423$ Comparison with 0.025 Do not reject H_0 Insufficient evidence that mean time in the library has changed	B1(AO1.1) B1(AO2.5) M1(AO3.3) A1(AO3.4) A1(AO1.1) M1(AO1.1) A1FT(AO2.2b) [7]	Must be stated in terms of parameter Values B1B0 for one error, e.g. undefined μ or 1-tail Stated or implied BC Allow comparison with 0.05 if $H_1 : \mu > 32.5$ In context, not definite; FT their 0.0423, but not comparison with 0.05	Use of 34.5 B0B0 OR $\frac{34.5 - 32.5}{8.2 \div \sqrt{50}}$ M1 allow without square root A1 = 1.725 A1 Comparison with 1.96 (allow comparison with 1.645 if $H_1 : \mu > 32.5$) FT their 1.725, but not comparison with 1.645
			Total	7		

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
9		a	$H_0 : p = 0.097$ $H_1 : p > 0.097$ where p is the proportion of patients experiencing side effects within a year $X \sim B(450, 0.097)$ and $X = 50$ $P(X \geq 51) = 1 - 0.862 = 0.138(3 \text{ s.f.})$ Comparison with 0.1 Do not reject H_0 No evidence (at 10% level) that proportion under new treatment greater than under standard treatment	B1 (AO1.1) B1 (AO2.5) M1 (AO3.3) A1 (AO3.4) A1 (AO1.1) M1 (AO1.1) A1 (AO2.2b) [7]	Must be stated in terms of parameters Undefined p B1B0 Stated or implied BC In context, not definite, e.g. Proportion not greater A0	Only 0.138 seen without parameters/distribution M1AO FT their 0.138, but not comparison with 0.1
		b	E.g. The patients could be treated together so they are not independent, so the binomial model is not valid. E.g. The 450 patients are not a random sample from the population, so the binomial model is not valid. E.g. It is not known whether the proportion of patients experiencing side effects under the standard treatment is 9.7%, so the binomial model used may not be valid.	B1 (AO3.5a) [1]	In context, referring to independence or random sampling. Must include a comment on appropriateness.	
			Total	8		